The light waves from different corresponding points meet at point P and interfere destructively to make it a minimum. The path difference δ between the waves from these corresponding points is, $\delta = \frac{a}{2}\sin\theta$

The condition for P to be first minimum is, $\frac{a}{2}\sin\theta = \frac{\lambda}{2}$

$$a \sin\theta = \lambda \qquad (7.37)$$

Condition for P to be second minimum

Let us divide the slit AB into four equal parts. Now, the width of each part is $a/4$. We have several corresponding points on the slit which are separated by the same width $a/4$. The path difference δ between the waves from these corresponding points is, $\delta = \frac{a}{4}\sin\theta$.

The condition for P to be second minimum is, $\frac{a}{4}\sin\theta = \frac{\lambda}{2}$

$$a \sin\theta = 2\lambda \qquad (7.38)$$

Condition for P to be third minimum

The same way the slit is divided in to six equal parts to explain the third minimum. The condition for P to be third minimum is, $\frac{a}{6}\sin\theta = \frac{\lambda}{2}$

$$a \sin\theta = 3\lambda \qquad (7.39)$$

Condition for P to be n^{th} minimum

Dividing the slit into $2n$ number of (even number of) equal parts makes the light produced by one of the corresponding points to be cancelled by its counterpart. Thus, the condition for n^{th} minimum is, $\frac{a}{2n}\sin\theta = \frac{\lambda}{2}$

$$a \sin\theta = n\lambda \qquad (7.40)$$

Where, n = 1, 2, 3 ... is the order of diffraction minimum.

Condition for maxima

For points of maxima, the slit is to be divided in to odd number of equal parts so that one part remains un-cancelled making the point P appear bright.

The condition for first maximum is,

$$\frac{a}{3}\sin\theta = \frac{\lambda}{2} \text{ (or) } a\sin\theta = \frac{3\lambda}{2} \qquad (7.41)$$

The condition for second maximum is,

$$\frac{a}{5}\sin\theta = \frac{\lambda}{2} \text{ (or) } a\sin\theta = \frac{5\lambda}{2} \qquad (7.42)$$

The condition for third maximum is,

$$\frac{a}{7}\sin\theta = \frac{\lambda}{2} \text{ (or) } a\sin\theta = \frac{7\lambda}{2} \qquad (7.43)$$

In the same way, condition for n^{th} maximum is,

$$a\sin\theta = (2n+1)\frac{\lambda}{2} \ (n^{\text{th}} \text{ maximum}) \qquad (7.44)$$

Where, n = 0, 1, 2, 3, . . . , is the order of diffraction maximum.

The central maximum is called 0th order maximum. The points of the maximum intensity lie nearly midway between the successive minima.

Note

Here, sin θ gives the angular spread of the diffraction from the central reference line. We can replace sin θ in the above equations with $\frac{y}{D}$. It is possible because θ is small. Now, we can approximate, sin θ = tan $\theta = \frac{y}{D}$

Where, y is the position of minimum (or) maximum on the screen from its center and D is the distance between the slit and the screen.

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